

ENGE702 · ENGINEERING MATHEMATICS III

Mathematics III Tutorial 2 — Worked Solutions

Fully worked, independently checked solutions with the original tutorial questions embedded beside the workings. Mathematics is pre-rendered as SVG, so symbols remain readable without MathJax, LaTeX, or an internet connection.

Worked solutions

Original questions embedded

Offline mathematics

PDF-ready

Scope and verification These are complete worked solutions to Questions 1–7 from the locally stored Tutorial 2 sheet. Question 8 is an instruction to continue with textbook exercises rather than a defined problem. The differential-equation residuals and initial conditions were checked symbolically; numerical results were independently recomputed.

Question 1 — Slope field, exact IVP solution, and Euler’s method

Tutorial 2

1. Consider the differential equation

$$y' - y = e^x.$$

- (a) Sketch the slope field for this differential equation. Plot on integer values of x and y between (and including) -2 and 2 .
(b) Sketch the solution to the initial value problem

$$y' - y = e^x, \quad y(-2) = 0$$

on your slope field.

- (c) Use Euler’s method to estimate $y(2)$, given $y(-2) = 0$. Use a step size of $h = 1$.

Original Tutorial 2, Question 1

The differential equation is

$$y' - y = e^x,$$

so the slope function is

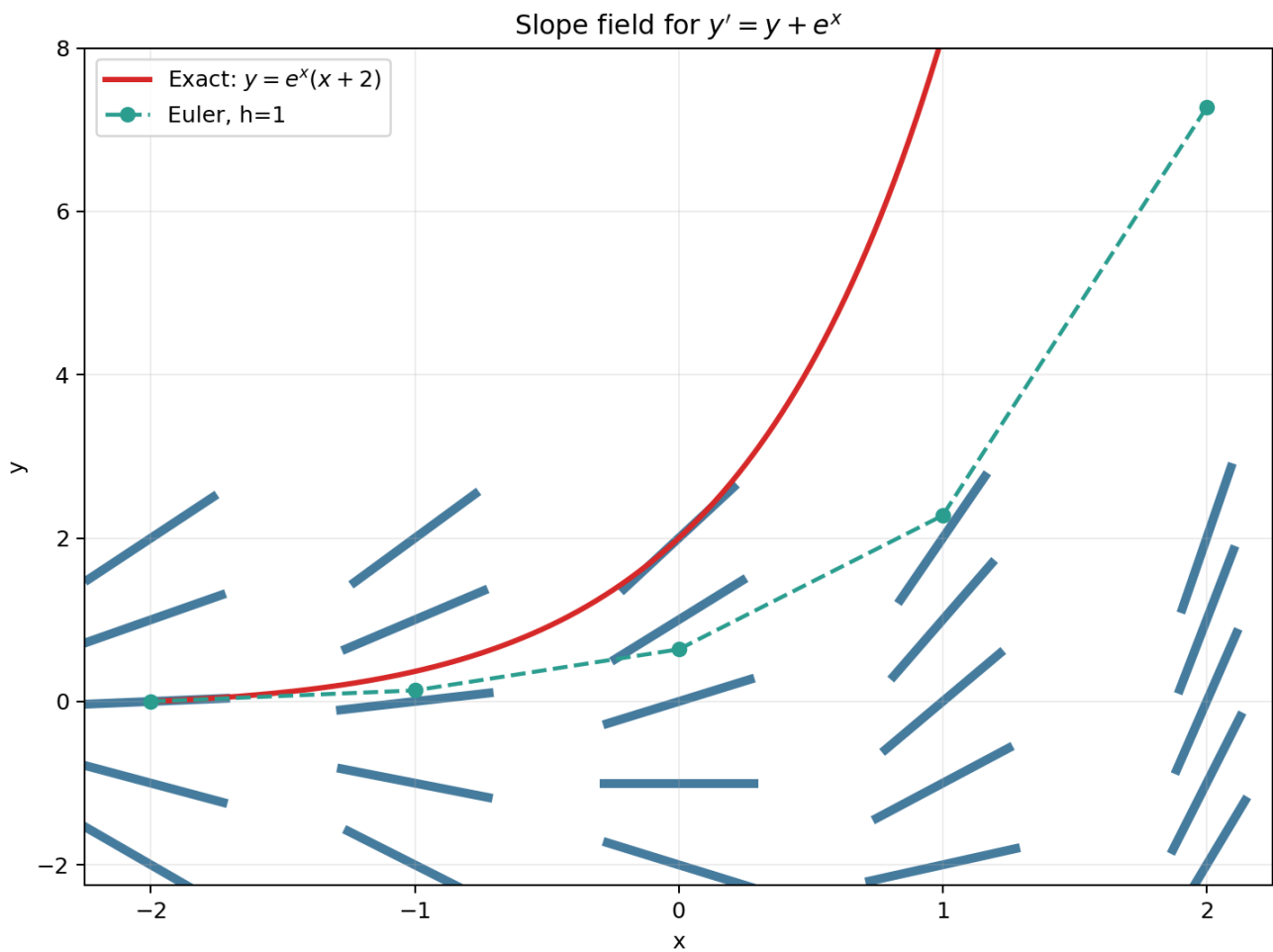
$$y' = f(x, y) = y + e^x.$$

(a) Slope field

At each integer grid point (x, y) , the short line segment has slope $y + e^x$. For example,

$$f(-2, 0) = e^{-2} \approx 0.1353, \quad f(0, 0) = 1, \quad f(2, 0) = e^2 \approx 7.389.$$

The field, exact IVP solution, and Euler polygon are shown below.



Slope field, exact solution, and Euler approximation

(b) Exact solution through $(-2, 0)$

Write the ODE in linear form:

$$y' - y = e^x.$$

The integrating factor is

$$\mu(x) = e^{\int -1 dx} = e^{-x}.$$

Multiplying through by e^{-x} ,

$$e^{-x}y' - e^{-x}y = 1,$$

and the left-hand side is a product derivative:

$$\frac{d}{dx}(e^{-x}y) = 1.$$

Integrating,

$$e^{-x}y = x + C,$$

so

$$y = e^x(x + C).$$

Apply $y(-2) = 0$:

$$0 = e^{-2}(-2 + C) \implies C = 2.$$

Therefore,

$$y(x) = e^x(x + 2).$$

(c) Euler estimate of $y(2)$, with $h = 1$

Euler's update is

$$y_{n+1} = y_n + h(y_n + e^{x_n}).$$

Starting from $(x_0, y_0) = (-2, 0)$:

n	x_n	y_n	$f(x_n, y_n) = y_n + e^{x_n}$	y_{n+1}
0	-2	0.000000	0.135335	0.135335
1	-1	0.135335	0.503215	0.638550
2	0	0.638550	1.638550	2.277100
3	1	2.277100	4.995382	7.272482

Thus,

$$y(2) \approx 7.27248 \text{ by Euler's method.}$$

For comparison, the exact value is

$$y(2) = 4e^2 \approx 29.5562.$$

The large difference reflects the coarse step size and the rapidly increasing solution.

Question 2 — Separable ODEs

- (c) Use Euler's method to estimate $y(2)$, given $y(-2) = 0$. Use a step size of $h = 1$.
2. Find the general solution to the following ODEs (note they are all separable):

(a) $y' = 2(y - 1)$

(c) $y' + xy = x$

(b) $y' - 4y = 0$

(d) $xy' = x^3y$

Original Tutorial 2, Question 2

(a) $y' = 2(y - 1)$

Separate variables:

$$\frac{dy}{y - 1} = 2 dx.$$

Integrate:

$$\ln |y - 1| = 2x + C.$$

Exponentiating and absorbing the sign into the constant,

$$y - 1 = Ce^{2x}.$$

Therefore,

$$\boxed{y = 1 + Ce^{2x}}.$$

(b) $y' - 4y = 0$

Since $y' = 4y$,

$$\frac{dy}{y} = 4 dx.$$

Hence

$$\ln |y| = 4x + C,$$

and

$$\boxed{y = Ce^{4x}}.$$

(c) $y' + xy = x$

Rearrange:

$$y' = x(1 - y).$$

Separate and integrate:

$$\begin{aligned}\frac{dy}{1 - y} &= x dx, \\ -\ln |1 - y| &= \frac{x^2}{2} + C.\end{aligned}$$

Therefore,

$$1 - y = Ce^{-x^2/2},$$

and, after renaming the arbitrary constant,

$$\boxed{y = 1 + Ce^{-x^2/2}}.$$

(d) $xy' = x^3y$

For $x \neq 0$, divide by x :

$$y' = x^2y.$$

Then

$$\frac{dy}{y} = x^2 dx,$$

so

$$\ln |y| = \frac{x^3}{3} + C.$$

Thus,

$$\boxed{y = Ce^{x^3/3}}.$$

The formula also includes the zero solution when $C = 0$.

Question 3 — First-order linear ODEs

$$(b) \quad y' - 4y = 0$$

$$(a) \quad xy' = x^2y$$

3. Find the general solution to the following ODEs:

$$(a) \quad y' = x + y$$

$$(c) \quad xy' + 4y = 2x$$

$$(b) \quad y' + 2y = x^2 + 4x + 7$$

$$(d) \quad xy' - 4y = x^5e^x$$

Original Tutorial 2, Question 3

(a) $y' = x + y$

Write

$$y' - y = x.$$

The complementary solution is $y_c = Ce^x$. Try a particular solution $y_p = ax + b$. Substitution gives

$$a - (ax + b) = x,$$

so $a = -1$ and $b = -1$. Therefore,

$$\boxed{y = Ce^x - x - 1}.$$

(b) $y' + 2y = x^2 + 4x + 7$

The complementary solution is $y_c = Ce^{-2x}$. Let

$$y_p = Ax^2 + Bx + D.$$

Then

$$y'_p + 2y_p = 2Ax + B + 2Ax^2 + 2Bx + 2D.$$

Match coefficients with $x^2 + 4x + 7$:

$$2A = 1, \quad 2A + 2B = 4, \quad B + 2D = 7.$$

This gives

$$A = \frac{1}{2}, \quad B = \frac{3}{2}, \quad D = \frac{11}{4}.$$

Hence,

$$\boxed{y = Ce^{-2x} + \frac{x^2}{2} + \frac{3x}{2} + \frac{11}{4}}.$$

(c) $xy' + 4y = 2x$

For $x \neq 0$, divide by x :

$$y' + \frac{4}{x}y = 2.$$

On any interval not crossing $x = 0$, an integrating factor is

$$\mu(x) = e^{\int 4/x \, dx} = x^4.$$

Therefore,

$$\frac{d}{dx}(x^4y) = 2x^4.$$

Integrating,

$$x^4 y = \frac{2x^5}{5} + C,$$

so

$$\boxed{y = \frac{2x}{5} + \frac{C}{x^4}}, \quad x \neq 0.$$

(d) $xy' - 4y = x^5 e^x$

For $x \neq 0$,

$$y' - \frac{4}{x}y = x^4 e^x.$$

The integrating factor is

$$\mu(x) = e^{\int -4/x \, dx} = x^{-4}.$$

Thus,

$$\frac{d}{dx}(x^{-4}y) = e^x.$$

Integrating,

$$x^{-4}y = e^x + C.$$

Therefore,

$$\boxed{y = x^4(e^x + C)}.$$

Question 4 — Initial-value problems

4. Solve the following initial value problems:

(a) $y' = yx^2$ $y(0) = 1$

(b) $xy' - xy^2 = 0$ $y(0) = 2$

(c) $y' - 2y = e^x$ $y(0) = 0$

(d) $xy' + y = x^2$ $y(1) = 0$

5. Water flowing through a hole in a tank obeys Torricelli's law, which says

Original Tutorial 2, Question 4

(a) $y' = yx^2$, $y(0) = 1$

Separate and integrate:

$$\frac{dy}{y} = x^2 dx,$$
$$\ln |y| = \frac{x^3}{3} + C.$$

Therefore $y = Ae^{x^3/3}$. Applying $y(0) = 1$ gives $A = 1$, hence

$$\boxed{y = e^{x^3/3}}.$$

(b) $xy' - xy^2 = 0$, $y(0) = 2$

For $x \neq 0$, divide by x :

$$y' = y^2.$$

Separate variables:

$$y^{-2} dy = dx.$$

Integrating,

$$-\frac{1}{y} = x + C.$$

Equivalently,

$$y = \frac{1}{K - x}.$$

Using $y(0) = 2$ gives $K = 1/2$. Therefore,

$$\boxed{y = \frac{2}{1 - 2x}}.$$

The solution is valid on the maximal interval containing the initial point, $x < 1/2$. Although the original equation degenerates at $x = 0$, the displayed function is differentiable there and satisfies the stated condition.

(c) $y' - 2y = e^x$, $y(0) = 0$

Use the integrating factor e^{-2x} :

$$\frac{d}{dx}(e^{-2x}y) = e^{-x}.$$

Integrating,

$$e^{-2x}y = -e^{-x} + C,$$

so

$$y = -e^x + Ce^{2x}.$$

Apply $y(0) = 0$: $0 = -1 + C$, hence $C = 1$. Therefore,

$$\boxed{y = e^{2x} - e^x}.$$

(d) $xy' + y = x^2$, $y(1) = 0$

Recognise a product derivative:

$$\frac{d}{dx}(xy) = x^2.$$

Integrating,

$$xy = \frac{x^3}{3} + C.$$

Thus

$$y = \frac{x^2}{3} + \frac{C}{x}.$$

Apply $y(1) = 0$:

$$0 = \frac{1}{3} + C,$$

so $C = -1/3$. Therefore,

$$\boxed{y = \frac{x^3 - 1}{3x}}, \quad x > 0.$$

Question 5 — Draining tank using Torricelli's law

5. Water flowing through a hole in a tank obeys Torricelli's law, which says

$$v(t) = 0.6\sqrt{2gh(t)},$$

where $v(t)$ is the velocity of the stream of water, $h(t)$ is the height of water above the hole, and $g = 9.81\text{m/s}^2$ is the gravitational constant.

Suppose we have a tank that has a diameter of 1m , a hole (at the bottom) with diameter 2cm , and an initial height of 3m .

Set up a model and solve to find a function describing the height of the water in the tank as a function of time.

Original Tutorial 2, Question 5

The tank radius is $R = 0.5\text{ m}$, so its cross-sectional area is

$$A_T = \pi R^2 = 0.25\pi \text{ m}^2.$$

The hole radius is $r = 0.01\text{ m}$, giving

$$A_h = \pi r^2 = 0.0001\pi \text{ m}^2.$$

The outflow speed is

$$v = 0.6\sqrt{2gh}.$$

Volume conservation gives

$$A_T \frac{dh}{dt} = -A_h v.$$

Therefore,

$$\frac{dh}{dt} = -0.6 \frac{A_h}{A_T} \sqrt{2g} \sqrt{h}.$$

Because

$$\frac{A_h}{A_T} = \frac{0.0001\pi}{0.25\pi} = 0.0004,$$

and $g = 9.81\text{ m/s}^2$, define

$$k = 0.6(0.0004)\sqrt{2(9.81)} = 0.001063067 \text{ m}^{1/2}/\text{s}.$$

The model is

$$\frac{dh}{dt} = -k\sqrt{h}, \quad h(0) = 3.$$

Separate and integrate:

$$\begin{aligned} \frac{dh}{\sqrt{h}} &= -k dt, \\ 2\sqrt{h} &= -kt + C. \end{aligned}$$

From $h(0) = 3$, $C = 2\sqrt{3}$. Hence

$$\sqrt{h} = \sqrt{3} - \frac{k}{2}t,$$

and the water height is

$$h(t) = \left(\sqrt{3} - 0.0005315336 t \right)^2 \text{ m}.$$

This applies until the tank empties. Setting $h = 0$,

$$t_{\text{empty}} = \frac{2\sqrt{3}}{k} \approx 3258.59 \text{ s} \approx 54.31 \text{ min}.$$

For later times the physical model is $h(t) = 0$.

Question 6 — RL electric circuit

rank as a function of time.

6. Suppose we have an electric circuit with a resistor of 1Ω , and inductor of $1H$ and an EMF given by $E(t) = \sin(t)$. Suppose at time $t = 0$ the current is $I = 0$. Find an expression for the current in the circuit as a function of time.
7. A particular machine component has a useful life of 10 years. Its value depreciates at

Original Tutorial 2, Question 6

Kirchhoff's voltage law for a series resistor-inductor circuit is

$$L\frac{dI}{dt} + RI = E(t).$$

With $L = 1\text{ H}$, $R = 1\Omega$, and $E(t) = \sin t$,

$$I' + I = \sin t.$$

Use integrating factor e^t :

$$\frac{d}{dt}(e^t I) = e^t \sin t.$$

Since

$$\int e^t \sin t \, dt = \frac{e^t}{2}(\sin t - \cos t),$$

we obtain

$$I = \frac{1}{2}(\sin t - \cos t) + Ce^{-t}.$$

Apply $I(0) = 0$:

$$0 = -\frac{1}{2} + C,$$

so $C = 1/2$. Therefore,

$$I(t) = \frac{1}{2}(\sin t - \cos t + e^{-t}) \text{ A}.$$

Question 7 — Depreciation model

7. A particular machine component has a useful life of 10 years. Its value depreciates at a rate proportional to the life of the product remaining. Suppose it is purchased for \$5000, and one year later its value is \$3800. Find a function that represents the value of the component as a function of time.

8. Finished all these? Work through problem sets 1.3 and 1.5 in the textbook

Original Tutorial 2, Question 7

Let $V(t)$ be the component value in dollars, with t measured in years after purchase. The remaining life is $10 - t$. Interpreting “depreciates at a rate proportional to the life remaining” gives

$$\frac{dV}{dt} = -k(10 - t), \quad k > 0.$$

Integrating,

$$V(t) = C - k \left(10t - \frac{t^2}{2} \right).$$

Since $V(0) = 5000$, $C = 5000$. Use $V(1) = 3800$:

$$3800 = 5000 - k \left(10 - \frac{1}{2} \right),$$

$$1200 = 9.5k,$$

$$k = \frac{2400}{19} \approx 126.3158.$$

Thus,

$$V(t) = 5000 - \frac{2400}{19} \left(10t - \frac{t^2}{2} \right).$$

An equivalent form is

$$V(t) = 5000 - \frac{1200}{19} (20t - t^2).$$

This is the mathematical result from the wording and the two supplied values. It predicts a negative value before the stated ten-year life ends, so a practical valuation model would impose a minimum salvage value or use a different depreciation law.

If the sheet intended to say that the depreciation rate is proportional to the **value remaining** rather than the **life remaining**, the standard exponential model would instead be

$$V(t) = 5000(0.76)^t,$$

because $V(1)/V(0) = 3800/5000 = 0.76$. The wording should therefore be confirmed with the lecturer; the boxed quadratic model above follows the text exactly.

Question 8 — Further study instruction

8. Finished all these? Work through problem sets 1.3 and 1.5 in the textbook

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Original Tutorial 2, Question 8

Question 8 does not provide additional equations to solve. It directs students who have completed the tutorial to work through textbook problem sets 1.3 and 1.5. Those textbook questions are not included in the local tutorial sheet, so no extra problems have been invented here.

Final answers at a glance

- **1(b):** $y = e^x(x + 2)$.
- **1(c):** Euler estimate $y(2) \approx 7.27248$ for $h = 1$.
- **2:** $1 + Ce^{2x}$, Ce^{4x} , $1 + Ce^{-x^2/2}$, $Ce^{x^3/3}$.
- **3:** $Ce^x - x - 1$; $Ce^{-2x} + x^2/2 + 3x/2 + 11/4$; $2x/5 + C/x^4$; $x^4(e^x + C)$.
- **4:** $e^{x^3/3}$; $2/(1 - 2x)$; $e^{2x} - e^x$; $(x^3 - 1)/(3x)$.
- **5:** $h(t) = (\sqrt{3} - 0.0005315336t)^2$ m, until $t \approx 3258.59$ s.
- **6:** $I(t) = \frac{1}{2}(\sin t - \cos t + e^{-t})$ A.
- **7:** $V(t) = 5000 - \frac{1200}{19}(20t - t^2)$.